

Preferential Transport of a Bromide Tracer Applied in a Pulse of Ponded Water

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ABSTRACT

The objective of this study was to quantify relations between preferential transport of a solute and initial water content, infiltration rate, and porosity in a field soil where preferential transport was mainly due to soil heterogeneity. We measured the horizontal and vertical distribution of a tracer chemical applied with ponded water to study the flow paths of the tracer. The soil at the site is a Bosville fine sandy loam (fine-mixed, thermic Albaquic Paleudalfs). Strontium bromide (SrBr_2) tracer was applied with a dye (methylene blue) in a 100 or 50-mm pulse of water to soil within eight double ring infiltrometers. After 48 h the soil in each infiltrometer was sampled to 0.7 m. Twelve horizontally oriented, continuous soil samples 0.1 m long were collected at each depth. There were very few dye stains of root hairs, root channels, and pores to a depth of about 50 to 80 mm. Recoveries of Br to 0.5 m ranged from 36 to 56% applied. Bromide recovery was negatively correlated with initial water content and positively with total porosity. Below 0.35 m in depth resident solute concentration at a sampling position was positively correlated with concentration in the layer above indicating preferential vertical flow paths. It appeared that a large fraction of solute transport was through the highly porous areas of the cross-section of soil bounded by the infiltrometer ring. The preferential transport of Br in this study was influenced largely by the properties of the clay layer at 0.35 m that had the lowest conductivity and lowest porosity in the profile and appeared to have cracks filled with sand.

WATER FLOW and solute transport in a soil can be described by a number of conceptual models. The most common is the convective-dispersive model where there is more or less complete displacement of antecedent water and dispersive spreading of the infiltrating front (Nielsen and Biggar, 1962). The convective-dispersive model is often applied to transport in soil columns or small field plots ($<1 \text{ m}^2$) because soil properties are often considered to be more horizontally uniform at this scale (compared with field scale).

When solute transport through a small area ($<1 \text{ m}^2$) diverges from the convective dispersive model it is termed *preferential transport* or *bypass flow*. This type of solute transport is distinguished from the spatial variability of solute transport on a field scale (over areas of 1 m^2 and larger) where the transport at each location is assumed to obey the convective dispersive principles (Biggar and Nielsen, 1976) and can be described by the stochastic stream-tube modeling of independent vertical soil columns (Dagan and Breslar, 1979; Jaynes et al., 1988; Jury and Roth, 1990). In studies done at the field scale, horizontal variation of solute concentrations are clearly seen and can be easily related to variations in

macroscopic soil properties (Butters and Jury, 1989; Jaynes et al., 1988).

Preferential transport has been shown to be caused by several different mechanisms. Bimodal pore size distributions and large differences in mobility of water between the interaggregate and intraaggregate pores of an otherwise horizontally homogeneous aggregated soil can cause preferential transport through larger pores (van Genuchten and Wierenga, 1976; Bond and Wierenga, 1990; Hart and Lowery, 1996). Another mechanism is the accelerated water flow through biomacropores—earthworm holes and root channels (Thomas and Phillips, 1979; White, 1985; Shipitalo, 1990; Singh and Kanwar, 1991; Heathman et al., 1995) and interpedal structural cracks in fine-textured soils (Dekker and Bouma, 1984; White et al., 1986). Several other researchers have shown fingering in coarse sands induced by instabilities in water flow (Glass et al., 1988; Hillel and Baker, 1988). Funneling of water flow caused by inclined lenses or deposition layers (Kung, 1993) may also result in preferential transport on a small scale. Preferential transport of solute has been found to be less pronounced in conventionally tilled soil than in no-till where the soil structure has not been disrupted (Andreini and Steenhuis, 1990; Bicki and Lei Guo, 1991).

Preferential flow may also be caused by lateral heterogeneities in soil texture or structure on a small scale, other than macropores. Lateral distributions of solute within the flow area on a small scale ($<1 \text{ m}^2$) are rarely investigated (Ellsworth and Boast, 1996). For this reason it is, generally, not possible to distinguish between preferential flow due to soil heterogeneities and preferential flow due to other mechanisms. The work of van Wesenbeck and Kachanoski (1994) and Ellsworth and Boast (1996) showed the existence of soil and transport heterogeneities on a small scale, and implied possible preferential transport.

Small-scale horizontal heterogeneities in soil texture and structure, other than macropores, could cause heterogeneity in soil porosity, initial water content, and hydraulic conductivity. Each or all of these horizontal heterogeneities combined could cause preferential flow and transport. If a soil is initially wet on the average, a large fraction of a surface-applied solute can be preferentially transported through a fraction of the soil that is better drained and has higher infiltrability (Jabro et al., 1991; Klodivko et al., 1991). In an initially dry soil, this preferential flow mechanism will be less pronounced (Bond and Wierenga, 1990) because of smaller differences in infiltrability among the laterally heterogeneous areas of the soil. The above results apply only to nonswelling soils. This phenomenon conforms to the concept of mobile-immobile regions in a homogeneous aggregated soil noted above, but on a larger scale.

Infiltration rates or hydraulic conductivities have been shown to be indicators of preferential transport. Jabro et al. (1991) found recovery of Br to be inversely

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related to final infiltration rate under ponded conditions. Jaynes et al. (1988) used infiltration rates to characterize the spatial variation of Br distribution on a field scale. Preferential transport as a result of lateral variation in hydraulic conductivity is more likely to occur during higher infiltration rates under ponded conditions than with lower infiltration rates during a rainfall (van Ommen et al., 1989).

Overall, although there has been some research in the area of preferential transport of solutes caused by small scale soil heterogeneities, we still do not have sufficient knowledge of soil parameters that control such a preferential transport. We have little knowledge of the lateral distributions of solute at this scale. Lateral mass exchange mechanisms are highly variable and can result in widely different transport regimes that are difficult to explain in terms of known state variables (Flühler et al., 1996). For this reason, there is also a need to investigate the suitability of specific soil parameters to be used as indicators of the potential for preferential solute transport.

The objective of this study was to further investigate the preferential transport and lateral distributions of solute that occur mainly due to soil heterogeneity, and to quantify the relations between preferential transport and initial water content, infiltration rates, and porosity in a field soil.

MATERIALS AND METHODS

The study site was at the USDA-ARS National Agricultural Water Quality Management Laboratory at Durant, OK. The soil at this site, a Bosville fine sandy loam (fine mixed, thermic Albaquic Paleudalf), had been under grass cover for at least 15 yr. A representative pedon of the Bosville series at this site has a very dark grayish brown fine sandy loam surface layer. The subsurface layer is dark brown fine sandy loam. The upper part of the subsoil is sandy clay loam; and the lower part is clay loam. The average depth distribution of these textural layers observed during sampling of the eight sites is given in Fig. 1 along with measured bulk densities. The soil profile at the site is underlain by fractured sandstone at about 0.65 to 0.70 m. Vertical sand lenses also appeared to be present in the clay loam layer at about 0.35 to 0.40 m in depth.

Measurement of Soil Hydraulic Properties

During the summer of 1989, double-ring infiltrometers were randomly located among eight locations, separated by a distance that varied from 3 to 5 m. The inner ring of the infiltrometer was 0.5 m in diameter and the outer ring was 0.9 m. The herbicide glyphosate [isopropylamine salt of N-(phosphonomethyl) glycine] was applied within six of the infiltrometers to kill the grass cover. Herbicide was not applied to Infiltrometers 1 and 5. Details of the measurement procedures were given in Timlin et al. (1994). In brief, one multiple-depth tensiometer was installed in the center of each infiltrometer by boring a hole with a soil auger slightly larger in diameter than the tensiometer. Before installation of the tensiometer, the hole was filled with a clay-silt slurry to minimize flow down the outside of the tensiometer tube. Each tensiometer contained five ceramic cups located at 0.1-m intervals to a depth of 0.5 m.

Ponded infiltration measurements were carried out in each of the eight infiltrometers. Saturated hydraulic conductivities

for the four layers (0–0.10, 0.10–0.20, 0.20–0.30, and 0.30–0.50 m) were calculated from the measured infiltration rates and the hydraulic gradients calculated from the tensiometer readings and depths.

After the ponded infiltration experiments were completed, the inner and outer rings were covered with plastic sheeting and hay to prevent evaporation and reduce temperature fluctuations. The infiltrometers were then covered with plywood sheets that were slotted to allow installation of the plywood around the tensiometers. Tensiometer readings were recorded during drainage until the drainage rate slowed to near zero, typically after 20 to 30 d. The tracer infiltration was conducted after drainage measurements were completed.

Following the tracer study (described in the next section), four replicate soil cores 54 mm in diameter and 60 mm in length were taken in each infiltrometer from four depths corresponding to observed soil horization described in Fig. 1. The horizons were differentiated by color and texture in the field as observed during sampling time. The sampling depths varied from infiltrometer to infiltrometer but were generally from 0 to 0.06, 0.1 to 0.16, 0.24 to 0.30, and 0.44 to 0.5 m. Moisture retention was determined on these cores (Klute, 1986). A sand table was used to determine volumetric water content at 0.3, 1.0, and 2.5 kPa soil water suctions relative to the midpoint of the core. A pressure plate apparatus was used for 5.0, 10.0, 20.0, and 50.0 kPa suctions. Moisture content at saturation and bulk density were also determined from these samples. The moisture release curve was used to obtain estimates of water content from tensiometric measurements of soil matric potential. Initial water content and porosity of the soil within the infiltrometers were calculated from these data.

Application and Sampling of Dye and Tracer

A dye, methylene blue ($C_{16}H_{18}ClN_3S \cdot H_2O$), was added to the 50 or 100-mm pulse of water at the rate of 0.2 g L^{-1} dye. The dye-solution was applied to the soil 1 to 12 d before the Br tracer was applied except for two of the infiltrometers

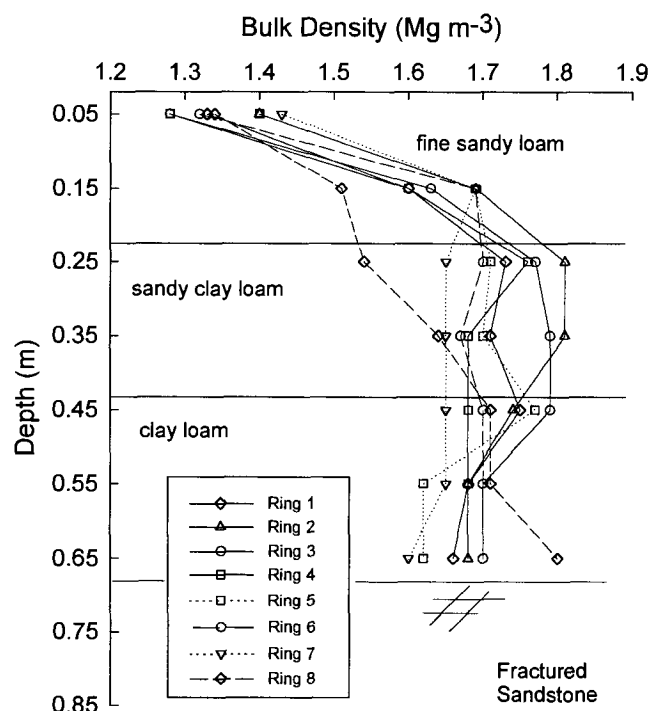


Fig. 1. Soil horization and bulk density with depth for soils within the eight infiltrometers in the study area.

Table 1. Tracer solution amount and concentration (C_0), and total porosity, initial relative saturation, and drained porosities to 0.50 m for soil within the eight infiltrmeters in the study.

Infiltrometer	Time between application of dye and application of tracer	Tracer		Total porosity	Initial relative saturation [†]	Drained porosity [‡]
		Tracer amount	C_0			
	d	mm	g m^{-3}	$\text{m}^3 \text{m}^{-3}$		$\text{m}^3 \text{m}^{-3}$
1	—§	50	30	0.369	0.539	0.261
2	1	50	30	0.310	0.653	0.233
3	7	50	30	0.334	0.638	0.260
4	10	50	30	0.306	0.664	0.252
5	1	100	21	0.312	0.622	0.205
6	5	100	21	0.332	0.780	0.250
7	3	100	21	0.352	0.520	0.266
8	5	100	21	0.380	0.543	0.253

[†] Proportion of total porosity filled with water before tracer application.

[‡] Mean water content for profile after 48 h of drainage.

§ No dye.

where the tracer and dye were applied together (Table 1). This application schedule was chosen to vary the initial water content, expressed in Table 1 as initial relative saturation, by allowing drainage to occur over varying periods of time.

Strontium bromide (SrBr_2) was used as a tracer chemical. The tracer was applied 1 to 2 mo after the conductivity and redistribution measurements. The infiltrmeters were divided into two groups. In the first group 30 g L^{-1} of tracer was applied in a 50-mm pulse of water, in the other 21 g L^{-1} was applied in a 100-mm pulse. Tracer was applied to both the inner and outer rings to minimize lateral flow (Ahuja et al., 1976). After application of tracer, the infiltrmeters were covered with plastic sheeting to reduce evaporation, and the solute and water allowed to redistribute for 48 h.

Core samples for Br analysis were taken in each infiltrmeter to 0.7 m in depth or to rock, whichever was deeper. The core diameter was 27 mm and the length was 100 mm. The 12 samples were relatively oriented in a constant spatial pattern (Fig. 2) to obtain samples continuous with depth in increments of 100 mm. This yielded from 72 to 84 samples per infiltrmeter location depending on depth sampled.

Dye observations were done at the same time as sampling for Br. The soil was scraped to expose stained macropores and root channels followed by coring samples for Br. In the surface 0.10 m we exposed soil at depth intervals of 20 mm for dye observation. The core samples for Br analysis in the top 0.10 m were also taken at 20-mm intervals. This was done

to allow us to observe dye staining before removing soil with the core. Before removing a layer, each hole made by the coring device was filled with a fine white sand as a marker to evenly expose the next layer 20 mm deeper, to align the next core samples vertically, and to prevent surface material from entering. The five sequential 20-mm interval samples taken from one location in the surface 0.1 m were composited for later analysis. Below 0.10 m the soil was exposed in 0.10 m thick layers for dye observations and to remove core samples.

Bromide was extracted from soil samples by mixing 20 g soil with 0.1 L distilled water (5:1 water/soil ratio), shaking for 30 min, decanting 0.05 L of the solution and centrifuging to separate the sediment. Bromide concentrations were determined using a bromide specific ion electrode (Orion Research, Boston, MA) (Onken et al., 1975; Adriano and Doner, 1982). Calibration of the bromide ion electrode was accomplished by using a standard solution of SrBr_2 .

Statistical Analysis

Statistical analysis for correlations were carried out using SAS software (SAS Inst., 1987). Because the tracer data were lognormally distributed we have reported the Spearman rank correlation coefficients.

RESULTS AND DISCUSSION

Dye Observations

Methylene blue is a very strongly adsorbed dye; therefore, the areas most strongly dyed would be surface-connected flow channels such as macropores or cracks. The dye will also stain the matrix if appreciable flow results in saturation of the sorbing sites.

There were very few dye stains on root hairs, in root channels, or in pores. The dye stains generally disappeared beyond 50 to 80 mm in depth. Within the upper 30 mm of soil some areas were more deeply stained than others, indicating appreciable localized matrix flow. Almost all the dye appeared to have been adsorbed by the soil matrix near the surface. In all plots, roots were plentiful to 180 to 200 mm, although there was negligible staining of root channels. Beyond this depth the root density noticeably decreased. Overall there was little evidence of significant macroporosity (i.e., visible stained or unstained macropores) or conducting root channels beyond 100 mm in depth.

This is in accordance with results reported by Timlin

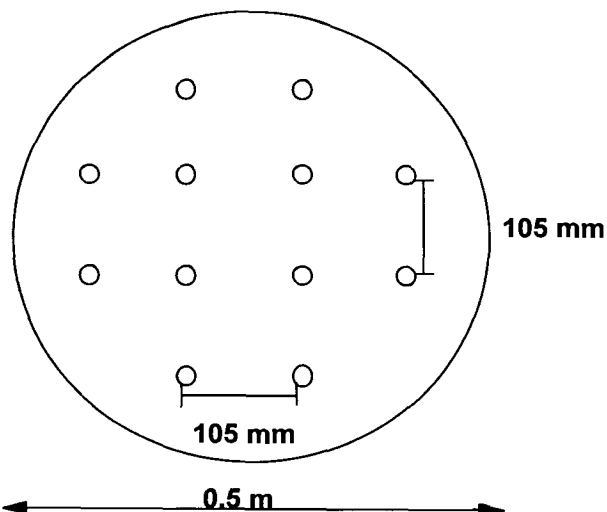


Fig. 2. Layout of Br sampling grid within the inner ring of the double ring infiltrmeter.

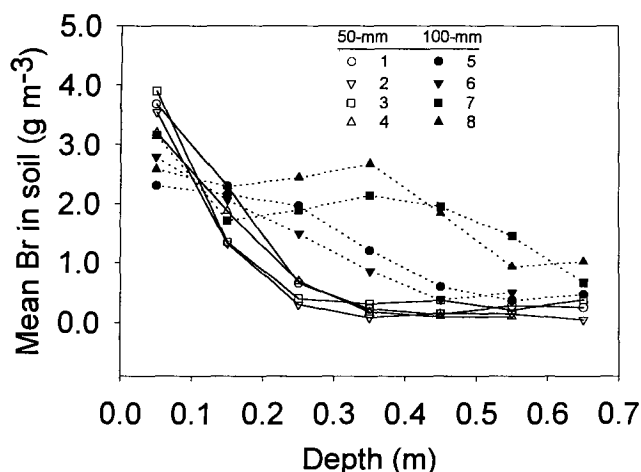


Fig. 3. Bromide content averaged by soil depth vs. soil depth for the soil within the eight infiltrmeters, four of which received a 50-mm tracer solution pulse and the other four a 100-mm pulse.

et al. (1994) where measured macroporosities were relatively small for the surface soil within the infiltrmeters and close to zero for the subsoil. It is not likely, therefore, that there were significant continuous macropores to influence solute movement.

Tracer Distribution

The vertical distributions of mean Br contents in different infiltrmeters are shown in Fig. 3 for the two application pulses. Each point represents the mean of 12 samples at a depth interval. Depths are reported as depths of the midpoint of a sampled interval, i.e., 0.15 m is the midpoint of the 0.10 to 0.20 m layer. The tracer in the infiltrmeters to which a 100-mm pulse was applied had penetrated deeper into the soil as compared with the tracer distribution for the 50-mm pulse. The deeper penetration for the 100-mm pulse was obviously due to the larger volume applied rather than the time required to infiltrate the pulse or the saturated hydraulic conductivity (Table 2). Among the 50-mm applications, for example, the solution in Infiltrmeters 1 and 4 penetrated deeper than the solution in Infiltrmeters 2 and 3, even though the values of K_{sat} for 1 and 4 in the depth range 0 to 0.20 m represent the largest (0.0325 mm s^{-1} for Infiltrometer 1) and smallest (0.0042 mm s^{-1} for Infiltrometer 4) values for the four locations receiving the 50-mm pulse.

Table 3. Mean, variance, and CV of Br over all sampling locations by depth and application.

Depth	Mean	Variance	CV
m	g m^{-3}	$(\text{g m}^{-3})^2$	%
50-mm pulse			
0–0.10	3.578	0.6314	22.2
0.10–0.20	1.721	0.3917	36.4
0.20–0.30	0.513	0.1269	69.4
0.30–0.40	0.203	0.0823	141.1
0.40–0.50	0.199	0.1289	180.5
0.50–0.60	0.198	0.0377	97.9
0.60–0.70	0.236	0.1732	176.6
100-mm pulse			
0–0.10	2.709	0.3990	23.3
0.10–0.20	2.058	0.2115	22.3
0.20–0.30	1.952	0.4947	36.0
0.30–0.40	1.723	1.1427	62.0
0.40–0.50	1.183	1.0133	85.1
0.50–0.60	0.713	0.5003	99.2
0.60–0.70	0.690	0.4647	98.8

The means and variances as a function of depth for the two application methods averaged over infiltrmeters are given in Table 3. The CVs increase with depth, generally because the means are decreasing without a similar decrease in variance. For the 50-mm application, the variances are highest for the depth interval from the surface to the 0.25-m depth. The variances are highest in the 0.35 to 0.55 m depth range for the 100-mm application. The highest variances occur in the region of the leading edge of the solute pulse for both the 50 and 100-mm applications. These results reflect the dispersion of solute at the location of the leading edge of the tracer pulse. This is similar to results of Ellsworth and Boast (1996), who reported highest CVs for NO_3^- in soil at the trailing and leading edges of the solute pulse.

The data show a strong vertical correlation structure. The Br concentration of samples taken from depths greater than 0.25 m are significantly correlated with Br concentration of samples at the same horizontal location at the depth just above (Fig. 4). The correlation is low and not significant ($P = 0.01$) at the surface. The correlations increase with depth from 0.25 to 0.55 m and are significant ($P = 0.01$). The correlations are highest for the samples from the 0.25 to 0.45 m depths (Fig. 4). These correlations show evidence of “stream tube” behavior in the soil where transport of Br is concentrated in specific locations. A spatial representation of the vertical correlation is shown in the three-dimensional plots of solute distribution in Fig. 5a and 5b from data in two representative infiltrmeters.

Table 2. Recovery of Br, time to infiltrate pulse, mean infiltration rate, and saturated hydraulic conductivity for the eight sites.

Infiltrometer	Recovery		Time to infiltrate pulse	Mean infiltration rate	K _{sat}		
	0.50 m	0.70 m			0–0.20 m	0.20–0.40 m	0.40–0.50 m
	%				mm s ⁻¹		
50-mm Pulse							
1	46.7	50.3	00:10	0.0833	0.0325	0.0061	0.0022
2	36.1	37.5	09:57	0.0014	0.0317	0.0039	0.0022
3	42.3	56.3	01:30	0.0093	0.0056	0.0047	0.0019
4	40.4	40.5	01:26	0.0097	0.0042	0.0022	0.0011
100-mm Pulse							
5	39.3	43.3	01:20	0.0208	0.0544	0.0131	0.0044
6	36.2	38.6	10:38	0.0026	0.0597	0.0086	0.0067
7	51.8	61.9	03:25	0.0081	0.0278	0.0133	0.0072
8	56.4	65.7	01:07	0.0249	0.0194	0.0114	0.0058

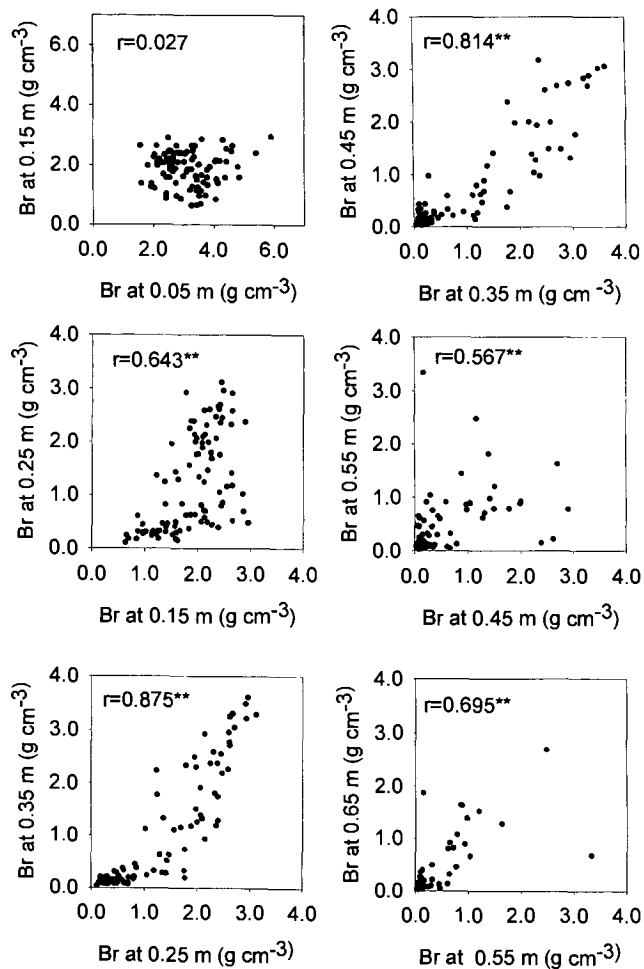


Fig. 4. Correlation of Br contents at neighboring depths at the same sampling grid location. Spearman rank correlation coefficients are given (** indicates $P < 0.01$).

The data in Fig. 4 and 5 suggest the possibility of preferential flow channels due to heterogeneities in the subsoil. The solute passed through the surface soil rather evenly, then channeled through sand filled cracks in the subsoil or other areas with porosity that was higher than the surrounding area. This is what Flüher et al. (1996) describe as "pushing flow toward funnel necks." As a result, a large portion of Br may have passed through the 0.7-m depth in only a small or partial cross-sectional area of a ring. Gupta et al. (1996) also reported little evidence of preferential transport in the surface horizons of a conventionally tilled soil. Dye distribution in the Ap and Bt1 horizons in the study of Gupta et al. (1996) was uniformly distributed but in the soil below these horizons only localized areas of the cross-sections of the soil cores were dyed. Van Wesenbeeck and Kachanoski (1994) reported that major lateral movement of solute in their study area began at the interface between the B and C horizons. Solute moved into tongues of B material that extended into the C horizon.

Bromide Recovery

Overall, the recoveries of Br are low (Table 2). These low recoveries could not be attributed to lateral flow

because tracer was also applied in the outer ring. It is difficult to obtain high mass recoveries in field soils where the lower boundary cannot be monitored. Jabro (1994) reported that 51% of Br applied in a 100-mm pulse of water leached below 0.80 m after 7 d. Flemming and Butters (1995) reported that mass recoveries of Br in a field study were high initially but decreased in later samples partially due to movement past the depth of sampling in some locations.

Displacement of soil water by an infiltrating solute-water pulse is not complete because of the effect of friction between the fluid and pore walls. As a result there can be significant dispersion of solute around a boundary defined by piston displacement even in a uniform soil (Nielsen and Biggar, 1962). We calculated a potential recovery for our experimental conditions as a comparison for the measured recoveries assuming some miscible displacement. The potential recovery was calculated by assuming that the infiltrating solution mixed completely with the water within the depth interval that would just hold the 50 or 100-mm volume of solution. This depth interval (called depth of infiltration) was calculated from the total and drainable porosities given in Table 1. Concentration of Br was calculated from the volume of water infiltrated, volume of antecedent water and Br concentration in the applied solution. Next a volume of drained water was calculated from the difference between total porosity and drained porosity (Table 1). The Br in this drained solution was assumed to be lost from the profile. The results of all calculations are shown in Table 4. The potential recoveries shown as percentages of the applied concentration are much higher than the measured recoveries (Table 4). The potential recoveries are conservative for the 0.50-m depth since recoveries are calculated for depths of approximately 0.20 m for the 50-mm pulse and 0.40 m for the 100-mm pulse (Table 3). This suggests that some mechanism of preferential flow contributed to solute transport in these soils.

Tracer Recovery and Initial Water Content

Recovery of Br within the upper 0.5 m was closely related to the percent of total pore space to 0.5 m that was filled with water (relative saturation) at the time of tracer application (Fig. 6). The recoveries were lowest in infiltrometers where the initial conditions were wet and highest where initial conditions were dry. When the soil is dry, large pressure gradients move water with solute into the small pores. Since these pores drain more slowly than larger pores, solute remains in the matrix during redistribution and vertical transport is reduced. Heathman et al. (1995) showed that dry aggregates at the surface of a soil column adsorbed a Br tracer and retarded its downward movement as compared with tracer application to cores without aggregates. As the proportion of pore space initially filled with water increases, relatively greater water movement takes place in larger and more continuous pore sequences because of the relatively greater velocities (Bond and Wierenga, 1990). Much of the tracer bypassed many of the water-

a

Infiltrometer 2, 50 mm pulse

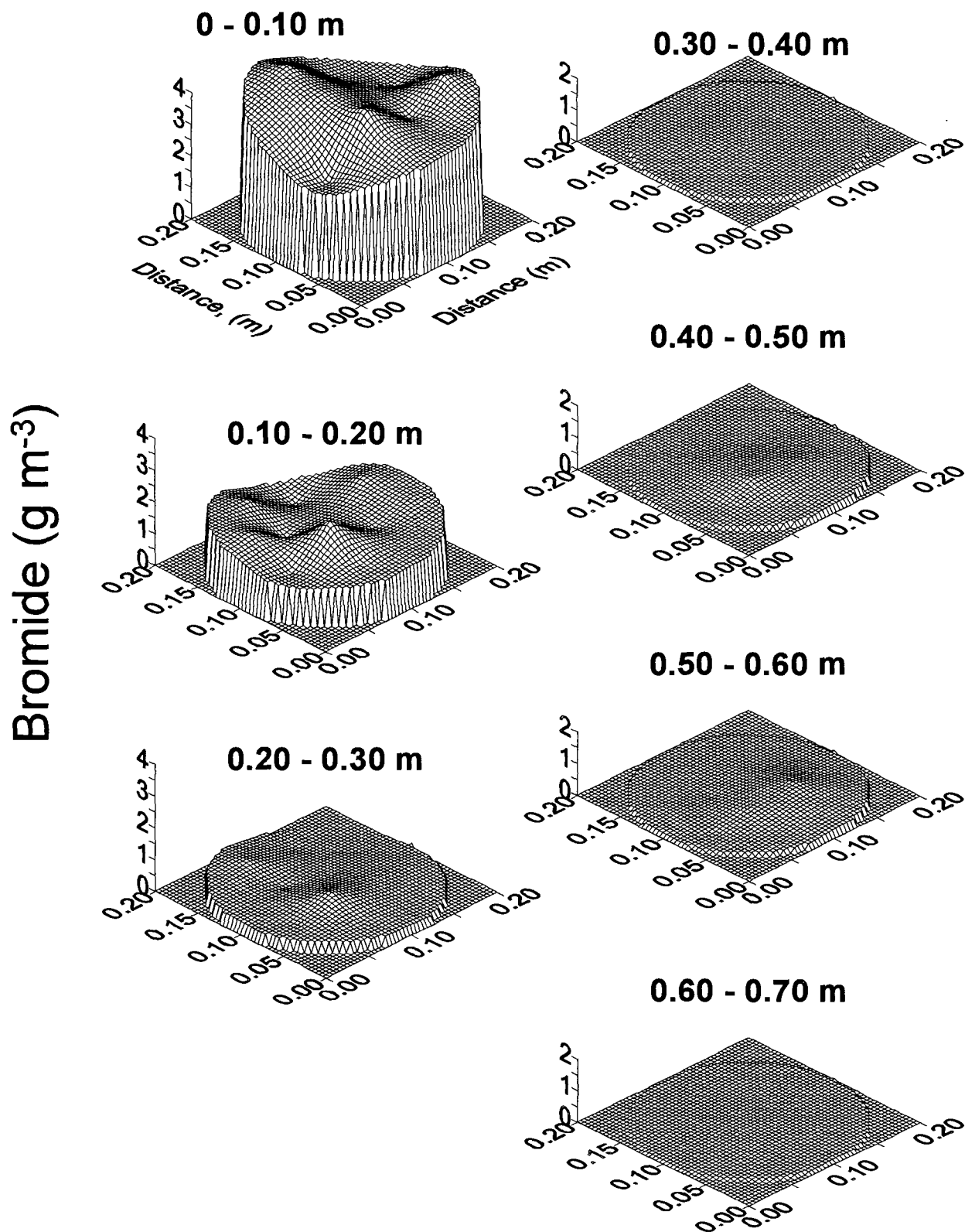


Fig. 5. Two-dimensional horizontal distributions of Br in the x and y directions with depth for two of the eight rings. (a) Ring 2, (b) Ring 6.
Fig. 5 cont. next page.

b

Infiltrrometer 6, 100 mm pulse

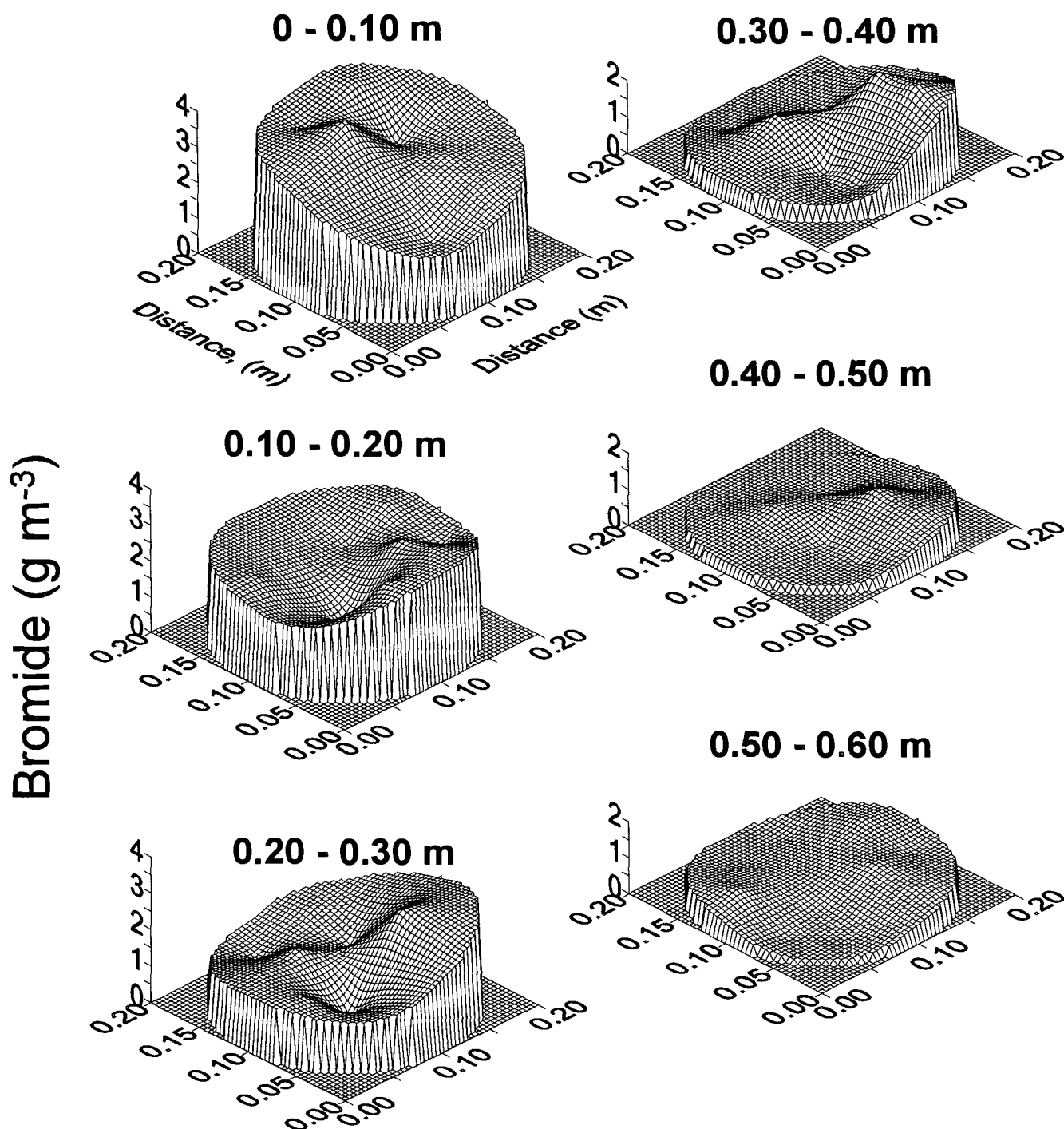


Fig. 5. Cont.

filled pores in the soil matrix during infiltration and drainage in our study site. The result was that recoveries of the tracer in the profile were correspondingly lower where a larger proportion of pore space was filled with water (wetter soil) at the time the solution was applied.

The effects of initial water content on solute concentration can be seen in Fig. 3. There was a large difference

in solute concentration and distribution with depth between the soils in Infiltrimeters 7 and 8 and the soils in Infiltrimeters 5 and 6 below 0.2 m in depth (Fig. 3). The soil in Infiltrimeters 7 and 8 had lower initial water contents than the soil in 5 and 6 (Table 1). We believe the relatively small amounts of Br recovered in the soil within Infiltrimeters 5 and 6 are partially due to dilution

Table 4. Potential recoveries of Br and their calculation.

Infiltrometer no.	Potential depth to which pulse will infiltrate	Volume of initial water	Relative concentration after mixing	Drained volume	Potential recovery to potential depth	Measured recovery to 0.50 m
	m	$\text{m}^3 \times 10^{-6}$		$\text{m}^3 \times 10^{-6}$	relative to applied	
1	0.136	2.7	0.65	1.5	0.81	0.47
2	0.161	3.3	0.61	1.2	0.85	0.36
3	0.150	3.2	0.61	1.1	0.87	0.42
4	0.163	3.3	0.60	0.9	0.89	0.40
5	0.321	6.2	0.62	3.4	0.79	0.39
6	0.301	7.8	0.56	2.5	0.86	0.36
7	0.284	5.2	0.66	2.4	0.84	0.52
8	0.263	5.4	0.65	3.3	0.78	0.56

of the infiltrating Br solution in the antecedent water. Some solute was also probably lost during redistribution and drainage. Since the initial water content was highest in the soils in Infiltrometers 5 and 6, there was greater potential in 5 and 6 for solute to have moved past the lower boundary during drainage and redistribution.

In the infiltrimeters where the 50-mm pulse was applied, the effect of initial water content on Br concentration is not as strong as for the 100-mm pulse. There were only small differences in initial water content for the soils in this group, however (Table 1). The soil in Infiltrometer 1 was the driest and had the highest Br concentration at 0.15-m depth (Fig. 3). Bromide recovery was also highest for this site among the four locations that received the 50-mm application (Table 1).

Tracer Recovery and Infiltration Rates

There was not enough data to show a significant correlation between recovery of Br and mean infiltration rate or measured K_{sat} . There was a tendency for the lowest values of mean infiltration rate (longest times to infiltrate the pulse) to be associated with the lowest recoveries, however (Table 2). Jabro et al. (1991) reported recovery of Br to be inversely related to final infiltration rate. In their study, however, macroporosity was a significant factor in the transport process.

There was no relationship between the mean infiltration rate and the measured hydraulic conductivity (Table 2). This may have been due to the swelling of the subsurface clay layer as well as the antecedent water

contents. The time to infiltrate the pulse was longest for Infiltrometers 2 and 6 (Table 2). Dye was applied with water to the soil in these infiltrimeters only 24 to 30 h before the solute pulse, so the clay layer may have been fully swollen at the time of tracer application. In the other infiltrimeters, dye was applied with the tracer or several days before (Table 1). Overall, these data suggest that the time to infiltrate the pulse was longest where the initial water contents were highest (Tables 1 and 2).

Tracer Recovery and Soil Porosity

Soils in the infiltrimeters with the lowest porosities within the upper 0.5 m had the lowest recoveries of Br (Fig. 7) and required more time to infiltrate the pulse (Table 2). This and the distribution of Br in Fig. 4 and 5 suggest that, in the most limiting soil layers (the clay loam layer and the fractured sandstone below), a large portion of flow was taking place in only small areas of the total cross-section within the infiltrimeters, most likely through sand filled cracks in the clay layer. Based on these results, we hypothesized that, when the overall porosity of the cross-section is high, proportionately more transport takes place through the highly permeable areas. Because flow is taking place through a larger area of the cross-section, relatively more antecedent water is displaced by the infiltrating solute pulse. This results in more retention of Br. This interpretation is similar to the interpretation of Bouma (1991), who had shown the bypass flow to depend on the ratio of hydrau-

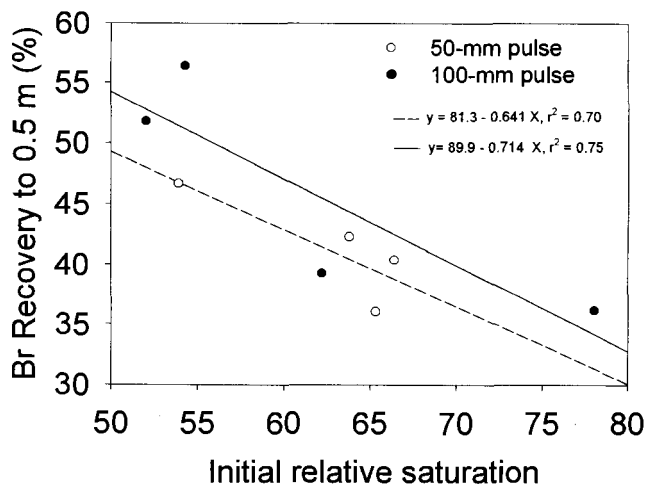


Fig. 6. Percent recovery of Br in 0 to 0.5 m vs. initial relative saturation (before tracer application).

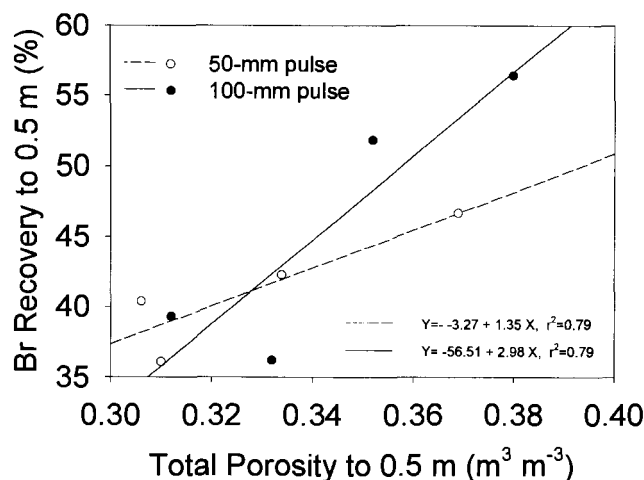


Fig. 7. Percent recovery of Br in 0 to 0.5 m vs. total porosity in the 0 to 0.5 m depth interval.

lic conductivity in the large and small pore fractions at a given water content.

Redistribution of Water and Tracer Recovery

Bromide recovery tends to be high where the increase in water in the profile is large between the time immediately before the pulse of tracer infiltrated and the time of sampling (Fig. 8). This relationship, shown in Fig. 8, is better defined for the 50-mm than for the 100-mm pulse. This is probably because a higher proportion of the applied water was retained in the profile where the 50-mm pulse was applied. The larger net increase in water for the infiltrometers given the 50-mm pulse could also be due to the lower porosities in these soils resulting in less drainage between tracer application and sampling (Table 1). The large negative change in water content in the upper 0.50 m between tracer application and sampling for Infiltrator 6 (Fig. 8) suggests that some of the tracer probably moved with redistributed water beyond the depth sampled for this site. This would have contributed to the loss of Br beyond 0.50 m.

CONCLUSIONS

In this study, recovery of Br from profiles with low porosity was less than the recovery from profiles with high porosity. We believe that this result arose from an interaction between mean permeability and permeability of localized regions within a cross-section of soil. The data suggest a large proportion of the Br was transported through localized regions of high porosity. Because the recoveries were highest in soils with the largest porosities, it appeared that relatively more Br moved through these localized regions of high porosity in the high porosity soils than in the low porosity ones. Because there was probably more transport through a larger fraction of the pore space in the high porosity soils, there was more complete contact with a wide range of pores resulting in more displacement and less bypass.

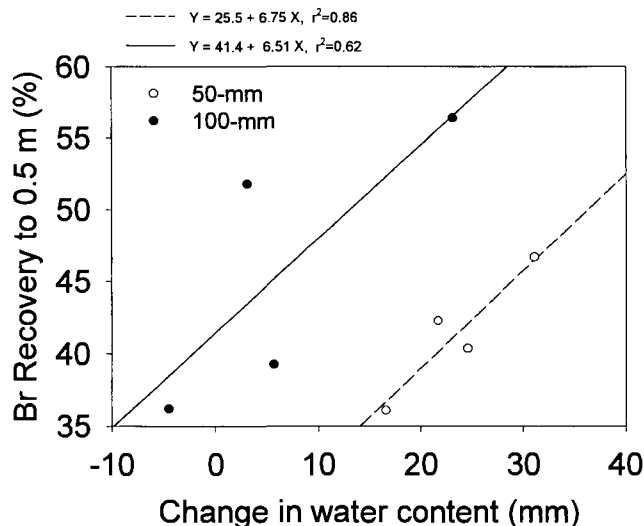


Fig. 8. Percent recovery of Br in 0 to 0.5 m vs. change in water in the 0 to 0.5 m depth interval over the time period from before tracer application to sampling.

Macropores, or other voids did not appear to be a major factor in the transport of Br in this study. There were very few stained pores or channels continuous to the surface and measured macroporosities were low.

The loss of Br from the profile was correlated with the antecedent water content, and relatively larger concentrations of Br were recovered from initially drier soils. This suggests that variations of water content as well as porosity can contribute to field-scale variability of solute concentrations.

The preferential transport of Br in this field study was influenced largely by the properties of the clay layer at 0.35 m that had the lowest conductivity and lowest porosity in the profile and appeared to have cracks filled with sand. Subsoil properties such as horizon thickness and tongues of A horizon material in the B horizon has been shown to affect solute dispersion (Ward et al., 1995). Knowledge of the properties of the entire soil profile are necessary to predict chemical transport; knowledge of properties at the soil surface is not sufficient.

Multidimensional models would be useful tools to use to study preferential transport in soils where transport is affected by variations in soil properties within a cross-section of soil. These types of models would be a useful tool to investigate the roles of porosity and hydraulic conductivity, and their interaction on solute transport.

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